

a) State any THREE roles of a system analyst.

1. Requirements Gathering – Collects and analyzes user requirements to understand what the system should do.
2. System Design – Translates business requirements into technical specifications for developers.
3. Liaison Role – Acts as a bridge between stakeholders (management, users, and developers) to ensure smooth communication and understanding.

b) Explain what prototyping is and explain TWO benefits of prototyping in software development.

Prototyping is the process of building a working model (prototype) of a system or part of it to visualize and refine user requirements.

Benefits:

Improved User Feedback – Users can interact with the prototype and provide feedback early in development.

Reduced Development Time and Costs – Issues can be identified and corrected early, minimizing the need for extensive changes later.

c) Explain the importance of using automated methods in fact-finding.

Automated methods such as surveys, software logs, and data analytics enhance accuracy and efficiency by collecting large volumes of data quickly and reducing human error.

d) State and explain any TWO types of systems.

Open System – Interacts with its environment by receiving inputs and providing outputs (e.g., a school system interacting with students and the community).

Closed System – Does not interact with its environment and operates in isolation (e.g., a chemical reaction in a sealed container).

e) State the conditions under which you would select an agile methodology.

Agile methodology is ideal when:

Requirements are expected to change frequently.

Client involvement is continuous and feedback-driven throughout development.

f) Explain any THREE benefits of performing system documentation.

Improves Maintenance – Helps developers understand system structure for easier troubleshooting and updates.

Supports Training – New users and developers can learn how the system works more quickly.

Enhances Communication – Provides a reference point for all stakeholders to understand system processes and design.

g) Explain THREE main categories of skills that a system analyst is expected to have.

Technical Skills – Knowledge of hardware, software, networks, and databases.

Analytical Skills – Ability to assess problems, identify solutions, and model system requirements.

Interpersonal Skills – Communication, teamwork, and negotiation skills to interact with users and stakeholders.

h) State and explain any THREE types of feasibility study.

Technical Feasibility – Assesses whether the current technology and resources can support the proposed system.

Economic Feasibility – Evaluates whether the benefits of the system outweigh the costs.

Operational Feasibility – Determines if the system will function well in the organization and be accepted by users.

i) State any FOUR differences between the two approaches of designing a system, that is, the system life cycle and the prototype.

System Life Cycle

Follows a sequential phase-by-phase process

Requirements are defined early

Time-consuming and rigid

Users are involved mainly at the beginning and end

Prototyping

Involves iterative development

Requirements evolve through user feedback

Faster development and more flexible

Users are continuously involved